Ocean Alkalinity Enhancement

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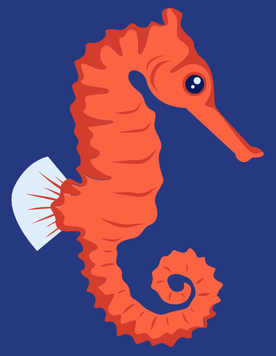
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Overview



What is Ocean Alkalinity Enhancement?

Ocean Alkanity:

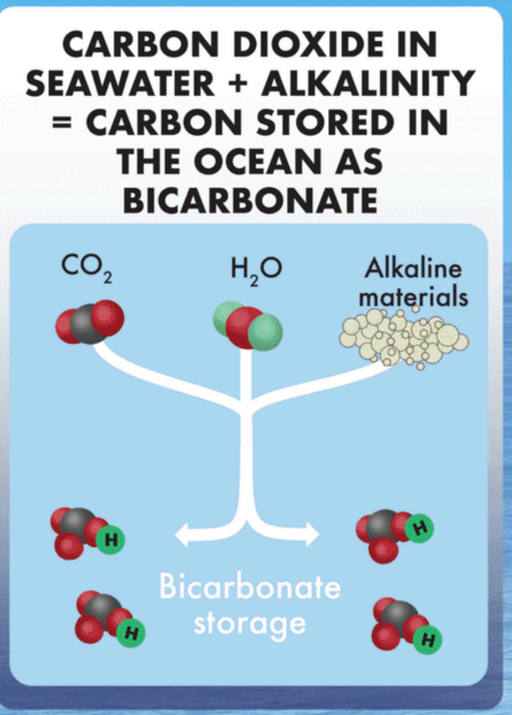
Ocean Alkanity is the seawater’s capacity to neautralize acids. Ocean Alkanity measures the concentrations of bases (ex. carbonates & bicarbonates) in seawater. These alkaline molecules react with hydrogran ions, which naturally raise the water’s PH and increases its ability to absorb CO2 from the atmosphere.



Ocean Alkanility Enhancement:

OAE is a carbon removal method that addes basic alkaline substances to seawater to speed up the ocean’s natural ability to lock away carbon dioxide. The three main methods for doing this are Mineral Dispersal, Electrochemical Systems, and Waste Systems.

OCEAN ALKALINITY ENHANCEMENT



Carbon dioxide

Alkaline solution returned to sea through outfall pipes

Coastal infrastructure can be co-purposed for alkalinity production and delivery

Alkalinity produced electrochemically from seawater

Seawater intake

Mining alkaline minerals

Silicates

Carbonates

Lime

Lime kiln

Beach spreading of silicate minerals

Ocean spreading of alkaline materials

CO_2 FROM LIME PRODUCTION SEQUESTERED UNDERGROUND

OCEAN VISIONS

oceanvisions.org/oceancdr

1. Mineral Disposal

Techniques

- Adds crushed silicate rocks, lime, or inductrial alkaine waste to seawater to convert CO2 into stable dissolved biocarbonate
- **Direct Disposal:** spreading finely ground minerals (ex. olivine or basalt) from ships into ocean surface waters.
- **Coastal Placement:** Placing alkaline sand or gravel on beaches to dissolve naturally via wave action
- **Industrial Waste Integration:** Uses alkaline leftovers from heavy industry to trap CO2 in the ocean (ex. mining waste, cement dust, etc)

Key Factors/ Challenges

- **Chemical Reaction:** Dissolved minerals raise seawater pH and try to pull in atmospheric CO2
- **Material Volume:** Isolating large amounts of carbon requires mining millions of tons of rock
- **Ecological Risks:** Unintended impurities in minerals or localized pH spikes can cause harm to marine life if not monitered closely.

2.Electrochemical Generation

Techniques

- Uses electricity to split seawater alkaline and acidic streams (contain basic compounds like sodium hydroxide that are returned to the ocean to absorb atmospheric CO2 while the acids are neutralized)
- **Seawater Splitting:** Electric currents divide dissolved salts and water molecules
- **Alkaline Production:** Creates a Base (ex. NaOH) that elevates local pH
- **Carbon Uptake:** Converts dissolved CO2 into stables biocarbonate and carbonate ions, pulling more CO2 gas out of atmosphere

Key Factors/ Challenges

- **Precision Control:** offers precise dosing compared to dumping raw crushed minerals (mineral disposal)
- **Energy Control:** Requires high electricity inputs, making project costs dependent on access to cheap renewable power

3.Carbon Conversion

Techniques

- Adds basic minerals or uses electricity to raise seawater pH. Converts dissolved CO2 into stable bicarbonate and carbonate ions
- **Electrochemical Splitting:** Electric current splits water or salt streams into acid and basic hydroxides. The base is added to the ocean while the acid byproduct is safely managed
- **Ionic Transformation:** Free alkaline molecules bind with hydrogran ions and dissolved CO2, permanently shifting carbon into stable forms that remain stored for thousands of years.

Key Factors/ Challenges

- **Atmospheric Drawdown:** Forces a net flux of CO2 from the air into the ocean surface to restore chemical balance
- **Acid Reduction**: Counteracts localized ocean acidifications, helping marine organisms live
- **Environmental Risks:** Potentional concerns about local pH shocks, impurities or heavy metals from crushed minerals, and high energy costs

02
Chemistry



Sometimes, reviewing concepts is a good idea

CO_2 Forms an Acid

Dissolved CO_2 makes a weak acid (carbonic acid), which partially dissociates into bicarbonate and carbonate ions, releasin H+

Alkalinity Buffers Acid

Total alkalinity (ocean's capacity to neutralize acid) is mostly driven by HCO_3^- and CO_3^{2-}. When you add alkalinity, you're adding species that consume H+ (bases)

Shifting Equilibrium

Adding alkalinity w/o adding carbon shifts equilibrium to the right, lowering seawater's partial pressure of CO_2. Lowering ocean pCO_2 makes ocean pull in more CO_2 From the atmosphere to re-equilibrate.

Sometimes, reviewing concepts is a good idea

Carbon Stays Locked Away

Once atmospheric CO_2 dissolves, it's chemically stable. This form of carbon can stay in the ocean for thousands of years instead of cycling back to the atmosphere.

Added Alkalinity

Sources like olivine or lime add base without a matching acid, and the imbalance raises total alkalinity.

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Geological Analogues



Geological Analogues

Definition:

An analogue = something you use as a comparison
Geological analogue = physical model, natural site, or mineral specimen used to represent and study complex Earth processes

What scientists look at when studying OAE:

Natural processes with the same chemistry, how fast they work, and what happened ecologically when they occurred at scale in Earth’s past

Examples

Silicate Weathering

Rain + CO_2 dissolves rock like basalt/olivine on land, releasing alkalinity that eventually reaches the ocean

Historical CO_2 drawdown

After the PETM (~56 million years ago), elevated silicate weathering helped pull the CO_2 spike back down over ~100-200 thousand years

Carbonate Compensation

When CO_2 rises, the ocean’s own carbonate burial slow, passively raising alkalinity over thousands of years

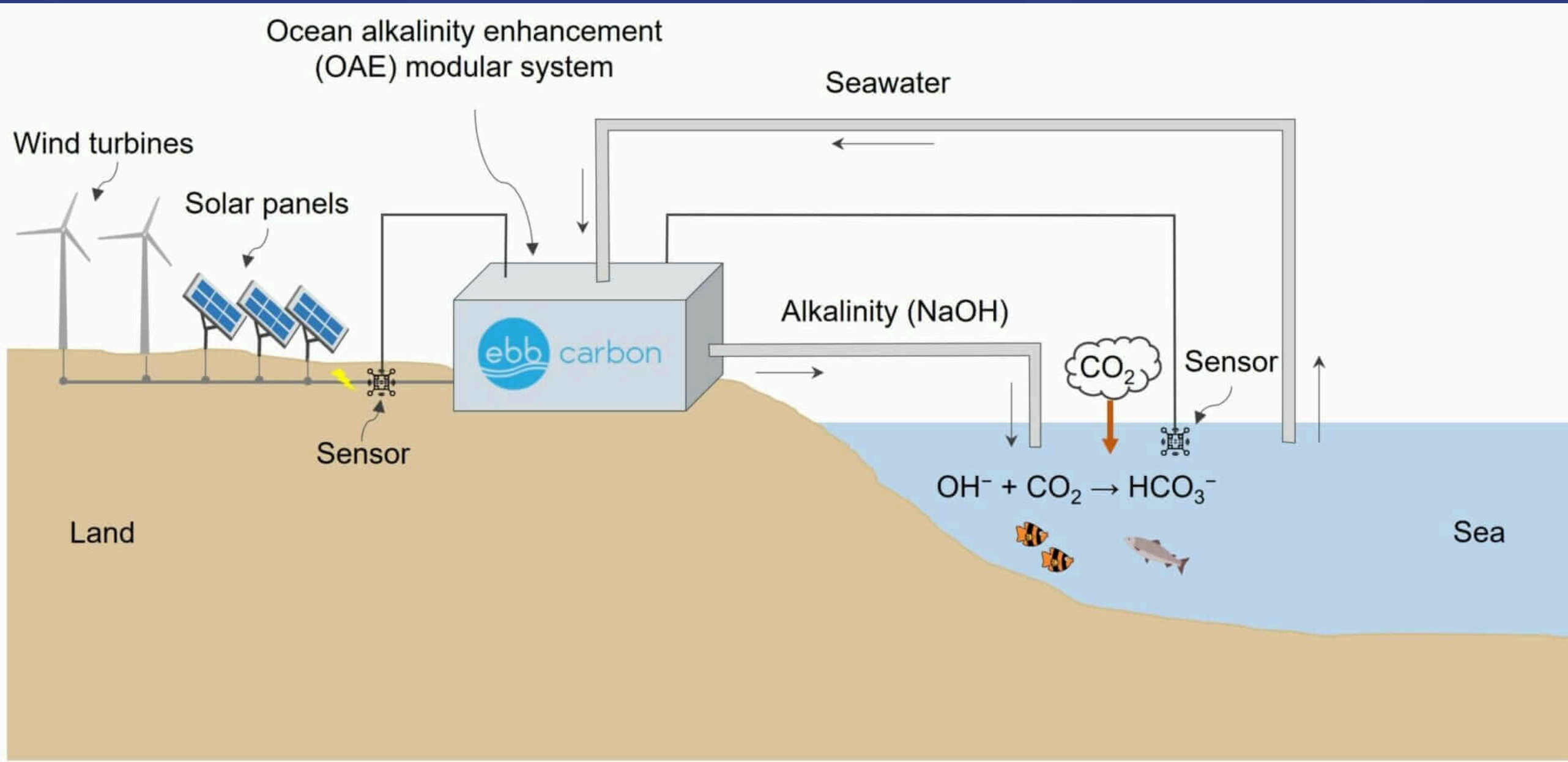
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Technology Development



Real World Demonstrations



- *Ebb Carbon* secured the first-ever permit (NPDES) to run an OAE experiment, and separately *Woods Hole Oceanographic Institution* got EPA approval for its own OAE experiment — both regulatory firsts
  - *Ebb:* Two-year project aiming to remove up to 1,000 tons of CO2 equivalent per year using electrochemical technology while mitigating local ocean acidification in the Wilkinson Basin in the Gulf of Maine
  - using ships and autonomous underwater vehicles to track how the alkalinity dispersed, presenting results at the Ocean Sciences Meeting 2026. They measured an extra 2–10 tonnes of CO_2 drawn into the ocean over a 4-day monitoring period — small, but a real measured result, not a model


Ocean alkalinity enhancement
(OAE) modular system
Seawater
Wind turbines
Solar panels
ebb carbon
Alkalinity (NaOH)
CO_2
Sensor
Sensor
$OH^- + CO_2 \rightarrow HCO_3^-$
Land
Sea

Monitoring Technology (Big Challenge!)



- **Monitoring, Reporting, and Verification (MRV)**
  - requires isolating small chemical changes in a constantly moving, open ocean
- Sensor Development
  - Total alkalinity (TA) sensors = most direct way to confirm whether a experiment is working but reliable, fully-working sensors are not commercially available
    - Underwater and surface robots follow the alkaline material after it's added so scientists can see where it goes
- Prediction Models
  - Scientists use computer modelling to predict how the alkaline material will spread through water so they know where to put sensors
    - Shared Database!!
      - (public database used to compare 5 different OAE projects)
- Key Challenges:
  - OAE disperses into moving, uncontained water so monitoring can't always be as accurate as one would like it to be
  - To get official results of carbon removal, proof of starting measurements, tracking as it dissolves, and full accounting fo emissions is required.

Scaling Technology



- First Carbon Credit!
  - Startup called *Planetary* got the world's first "carbon credits" for OAE (proof that a trusted system now existed to verify and reward the OAE experiments)
  - were able to safely pull carbon dioxide from the air, lock it into seawater as stable bicarbonate, and lower local ocean acidity
- Government Approval
  - Two companes (Ebb Carbon and WHOI) became the first ever to get offical government permits to test OAE
    - opens legal path for others to follow
- Bigger Delivery Systems
  - *Planterary* signed a $31 million dollar deal to massively scale up how much alkaline material they can deliver from 2026-2030
    - small tests moving to real-world size
    - making real impact
- Key Challenges
  - The minerals requires mining and shipping millions of tons of rock
    - scaling up = builing supply chain, not just doing more chemistry and experiments

What *Planetary Technologies* does!

1

Alkalinity, in mineral form like magnesium oxide and calcium oxide, is added to coastal waters in controlled amounts. Planetary carefully measures how much is added to ensure it stays within safe limits for the ecosystem.

2

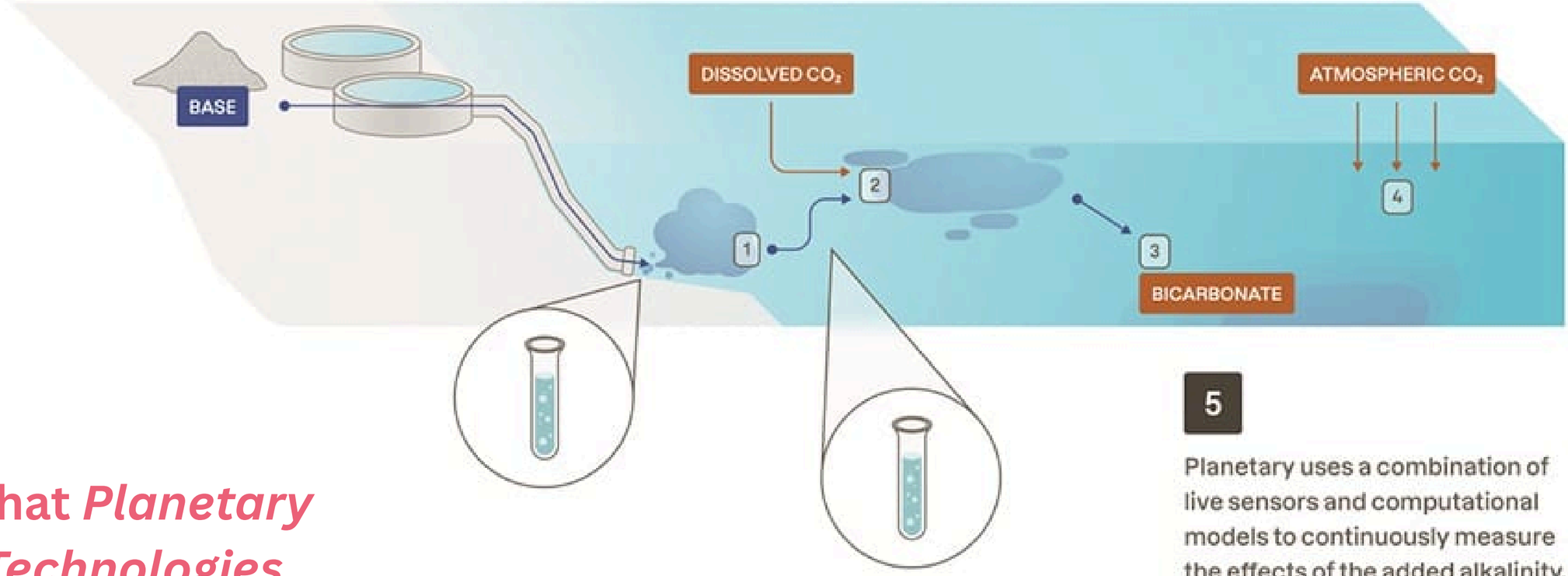
This added alkalinity helps neutralize carbon dioxide and carbonic acid in the water and locally reverses acidification.

3

The alkalinity stabilizes CO_2 in the form of the neutral salts bicarbonate and carbonate, which are already common in the ocean and are stable for around 10,000 years.

4

By removing dissolved CO_2 from the water, space is freed up for more CO_2 from the air to be absorbed into the ocean. This process is known as "air-sea-gas exchange" and results in atmospheric carbon removal.



5

Planetary uses a combination of live sensors and computational models to continuously measure the effects of the added alkalinity, to quantify how much CO_2 is removed, and to prevent any harm from the ecosystem.

05
Feasibility



Cost/Scale

Cost

- Right now, OAE costs roughly **$250-850** per tons of CO2 removed because technology is still at its early stages
- *Planetary's* mineral-based approach costs ~$270/ton, while other electrochemical method run closer to ~$500/ton
- Experts are projecting that cost could eventually drop to $50-$160/ton which would make this one of the cheapest carbon removal methods available!
    - trending in the right direction

Scale

- Theoretically, OAE could remove several billion tons of CO2 per year
    - enough to matter for global climate goals
- However, by far only 730 tons **total** have been verified worldwide to be removed via OAE
    - only a tiny fraction of potenial...

Offical Assessment/Key Challenges

Offical Assessment

- A *National Academics (NASEM)* reports rated OAE as having high confidence in its effectiveness
  - However, flagged as medium environmental risk and medium-to-high scale-up costs
- Report recommended $125-200 million in additional research funding before large-scale rollout, especically to study impacts on marine life
- Clearly more work needs to be done as OAE is not nearly fully developed but is progressing in the right direction

Key Challenges

- Profitability depends heavily on carbon credit prices
  - recent economic modeling found OAE plants are only profitable under high carbon-price scenarios
- Essentially, the chemistry of OAE works but more research and funding is required before this method is able to ight climate change at a high scale.

Citations

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Thank you!

